

Ethics Report for Case 28: Artificial Intelligence: Tap or Slam the Brakes

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Abstract— Artificial intelligence (AI) technologies are increasingly being adopted across industries, particularly in law enforcement and national security applications. Case Study 28 examines Boundless Brain, a startup developing AI-powered facial recognition systems integrated into police body cameras and surveillance equipment. While the technology promises enhanced public safety and crime prevention, it also raises serious ethical and professional concerns related to accuracy, bias, privacy, and civil liberties. This report analyzes the ethical and professional challenges presented in the case, with particular attention to the responsibilities of engineers under the IEEE Code of Ethics. Relevant research on facial recognition ethics and responsible AI standards is reviewed to establish context. The analysis highlights safety risks, legal considerations, and the disproportionate impact on marginalized communities. Finally, the report discusses how adherence to ethical standards and professional responsibilities can guide engineers toward responsible solutions and support long-term professional development.

Keywords— Artificial Intelligence, Facial Recognition, Engineering Ethics, IEEE Code of Ethics, Privacy, Bias

I. INTRODUCTION

Artificial intelligence has rapidly become a foundational technology in modern engineering, enabling systems that can analyze data, recognize patterns, and make decisions with minimal human intervention. One of the most controversial applications of AI is facial recognition technology, particularly when used by law enforcement agencies. Case Study 28 focuses on Boundless Brain, a private AI startup seeking to integrate facial recognition algorithms into police body cameras and surveillance systems for real-time identification of suspects.

This case is highly relevant to engineering ethics because it demonstrates the tension between technological innovation and ethical responsibility. While the technology could improve public safety and efficiency, it also presents risks related to inaccurate identification, bias, privacy invasion, and misuse by authorities. Prior research has shown that facial recognition systems often perform unevenly across demographic groups, which raises serious concerns about fairness and discrimination [1].

This report begins with background research on facial recognition ethics and AI standards. It then examines the professional and ethical issues present in the case, connects them to the IEEE Code of Ethics, and concludes with reflections on how engineers can apply these lessons in future professional practice.

II. BACKGROUND INFORMATION

Facial recognition technology (FRT) is a subset of artificial intelligence that uses machine learning algorithms to identify or verify individuals based on facial features. While FRT has been adopted for applications such as smartphone security and airport screening, its use in law enforcement has generated significant ethical debate. Studies indicate that facial recognition systems can exhibit higher error rates for women and people of color, leading to concerns about bias and unequal treatment under the law [1].

Ethical analyses of facial recognition emphasize the importance of transparency, accuracy, and accountability in system design and deployment. Roundtree's content analysis of facial recognition codes of ethics highlights that many ethical frameworks call for disclosure of system limitations and proactive efforts to mitigate bias before deployment [1]. Despite these recommendations, companies often move forward with real-world implementation without fully addressing these risks.

Additionally, the absence of comprehensive regulation places greater responsibility on engineers and organizations to self-govern AI technologies. Initiatives such as the Center for Standards and Ethics in Artificial Intelligence stress the importance of collaboration among industry, government, and academia to ensure AI systems are safe, reliable, and socially responsible [2]. These principles directly relate to the Boundless Brain case, where leadership disagrees on whether regulation should precede innovation.

III. PROFESSIONAL ISSUES

Several professional issues are present in this case, including teamwork, communication, safety, legal compliance, and adherence to standards. Within Boundless Brain, there is a clear

breakdown in internal communication between company leadership and employees. Engineers raised concerns about the technology's high error rate and potential misuse, yet leadership continued pursuing partnerships with law enforcement agencies.

From a professional standpoint, engineers are expected to engage in open communication and address technical concerns raised by team members. Ignoring internal criticism undermines teamwork and weakens ethical decision-making. This situation also reflects a conflict between business interests and professional responsibility.

Safety and legal compliance are also major professional concerns. Deploying a facial recognition system with a reported 35% error rate in real-time law enforcement settings poses serious risks. Misidentification could lead to wrongful arrests, violations of constitutional rights, and loss of public trust. Engineers have a professional obligation to ensure systems meet acceptable safety and reliability standards before deployment.

Industry standards for responsible AI stress the importance of testing, transparency, and accountability throughout a system's lifecycle [2]. Applying these standards correctly is essential for ethical and professional engineering practice, especially in high-risk applications such as surveillance.

IV. ETHICAL ISSUES

The Boundless Brain case presents multiple ethical dilemmas directly related to the IEEE Code of Ethics. One major concern involves IEEE Code 1, which requires engineers to prioritize the safety, health, and welfare of the public. Deploying facial recognition technology with known accuracy limitations places individuals at risk of harm through misidentification and unjust treatment.

IEEE Code 3 emphasizes honesty and realism in stating technical claims. Promoting AI surveillance tools while minimizing known limitations violates this principle. Engineers have an ethical responsibility to clearly communicate system capabilities and risks.

IEEE Code 5 requires engineers to improve public understanding of technology and its consequences. In this case, limited transparency regarding how facial recognition data is collected and used prevents informed public discussion and consent.

Finally, IEEE Code 8 mandates fair treatment of all individuals regardless of race or gender. Research indicates that facial recognition systems often perform poorly for marginalized groups [1]. Allowing such biased systems to be used in law enforcement contexts conflicts with ethical engineering responsibilities and contributes to social injustice.

V. CONCLUSION

Case Study 28 highlights the ethical and professional challenges that arise when emerging technologies are deployed without sufficient safeguards. While AI-powered facial recognition systems may offer benefits for public safety, premature deployment introduces serious risks related to bias, privacy, and civil liberties.

Engineers play a critical role in addressing these challenges by adhering to professional standards and the IEEE Code of Ethics. By prioritizing public welfare, transparency, and fairness, engineers can help ensure that AI technologies are developed responsibly. The lessons learned from this case emphasize the importance of ethical decision-making and its impact on long-term professional practice.

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